



EXAM MEMO

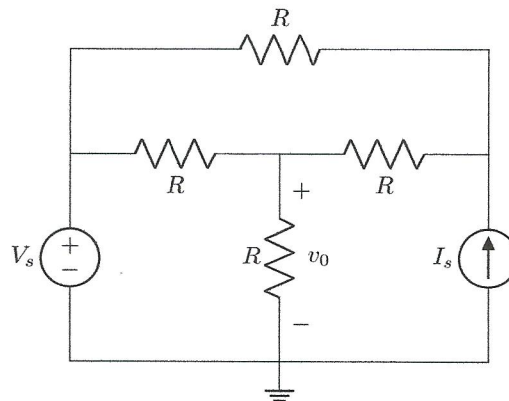
June 2020

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2020EBN111E01

Section 1

Questions 1 to 2 [4]



- 01** Determine the contribution to v_0 of the voltage source V_s alone if $I_s = 1 \text{ mA}$ and $R = 6 \text{ k}\Omega$. [2]
(The value of V_s is given in ClickUP.)
- 02** Determine the contribution to v_0 of the current source I_s alone if $V_s = 10 \text{ V}$. [2]
(The values of I_s and R are given in ClickUP.)

01

$V_s = 10 \text{ V}$

$(6+6) \parallel 6 = 4 \text{ k}\Omega$

$V_s = 10 \text{ V}$

$V_0 = V_s \left(\frac{6 \text{ k}}{6 \text{ k} + 4 \text{ k}} \right) = \frac{3}{5} V_s [\text{V}] (*)$

$= \frac{3}{5} (10) = 6 \text{ V}$

02

$R = 6 \text{ k}\Omega$

$I_s = 1 \text{ mA}$

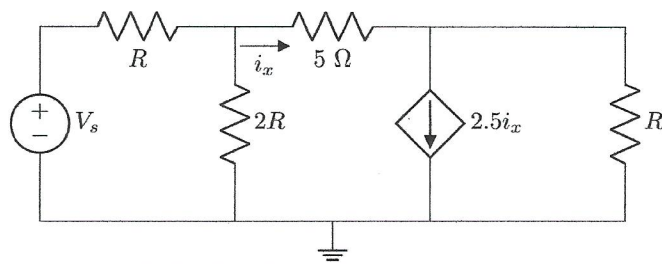
$R = 6 \text{ k}\Omega$

$R \parallel R$

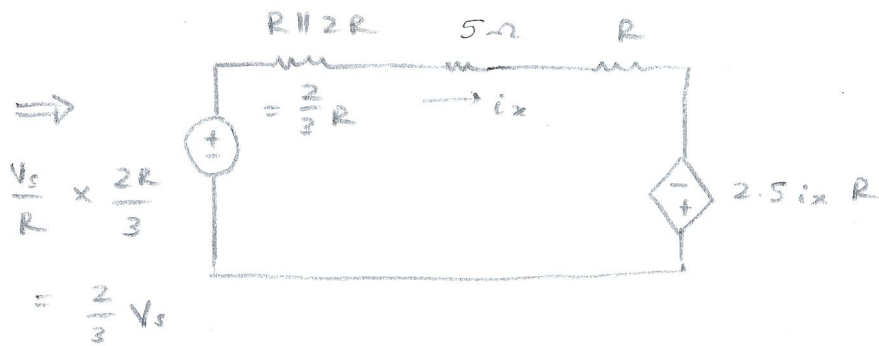
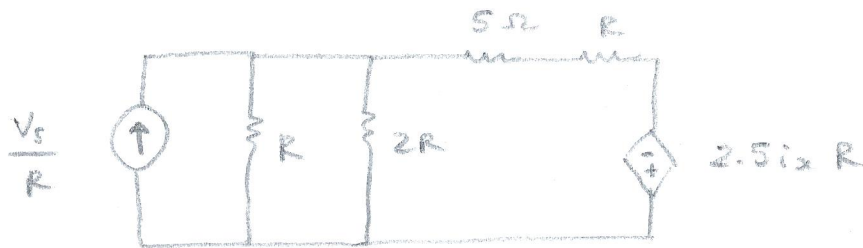
$i_0 = I_s \left(\frac{R}{R + \left(\frac{R}{2} + R \right)} \right) = \frac{I_s R}{\frac{5}{2} R} = \frac{2}{5} I_s$

$V_0 = \frac{2}{5} I_s \times \frac{R}{2} = \frac{I_s R}{5} [\text{V}] (*) = \frac{(1 \text{ mA})(6 \text{ k}\Omega)}{5}$

Question 3 [2]



- 03 Use source transformation to determine i_x . [2]
(The values of V_s and R are given in ClickUP.)



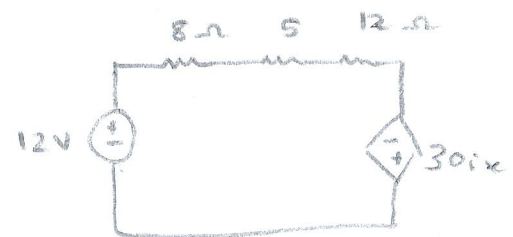
$$i_x = \frac{\frac{2}{3}V_s + 2.5i_x R}{\frac{2}{3}R + 5 + R}$$

$$i_x = \frac{\frac{2}{3}V_s}{(5 - \frac{5}{6}R)} \quad [A] \quad *$$

$$V_s = 18V, \quad R = 12\Omega:$$

$$I = \frac{18}{12} = 1.5A$$

$$12 \parallel 24 = 8\Omega$$



$$i_x = \frac{12 + 30i_x}{25}$$

$$25i_x = 12 + 30i_x$$

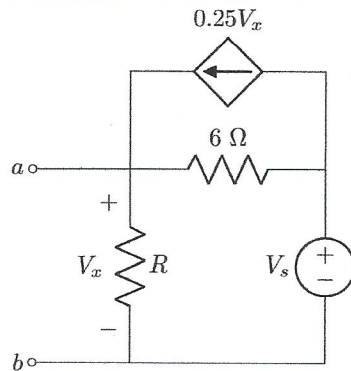
$$i_x = -\frac{12}{5}$$

$$\therefore i_x = -2.4A$$

Questions 4 to 5 [4]

Find the Thevenin equivalent voltage V_{TH} and Thevenin equivalent resistance R_{TH} , with reference to terminals $a - b$.

(The values of V_s and R are given in ClickUP.)



04 Determine V_{TH} . [2]

05 Determine R_{TH} . [2]

04

$$V_{TH} = V_x$$

$$V_s = 45 \text{ V}$$

$$R = 4 \Omega$$

$$\frac{V_x}{R} + \frac{V_x - V_s}{6} = 0.25 V_x$$

$$\frac{V_x}{4} + \frac{V_x - 45}{6} = 0.25 V_x$$

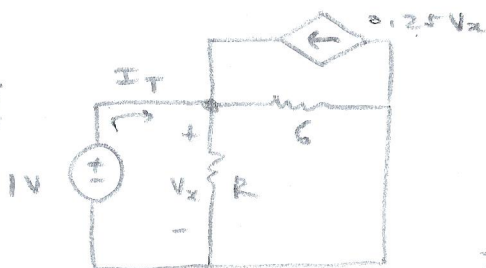
$$V_{TH} = \frac{V_s R}{6 - 0.5 R} \text{ [V]} (*)$$

$$(x12): 3V_x + 2V_x - 90 = 3V_x$$

$$2V_x = 90$$

$$V_x = 45 \text{ V}$$

05



$$V_x = 1 \text{ V}$$

$$I_T = \frac{1}{R} + \frac{1}{6} - 0.25(1)$$

$$R = 3 \Omega$$

$$R_{TH} = \frac{1 \text{ V}}{I_T} = \frac{1}{\left(\frac{1}{R} + \frac{1}{6} - 0.25\right)}$$

$$R_{TH} = \frac{1}{\frac{1}{3} + \frac{1}{6} - 0.25}$$

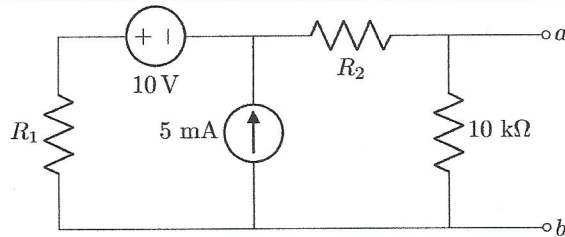
$$x6R: = \frac{6R}{6 - 0.5R} \text{ [\Omega]} (*)$$

$$R_{TH} = 4 \Omega$$

Questions 6 to 7 [4]

Find the Norton equivalent current I_N and Norton equivalent resistance R_N , with reference to terminals $a - b$.

(The values of R_1 and R_2 are given in ClickUP.)

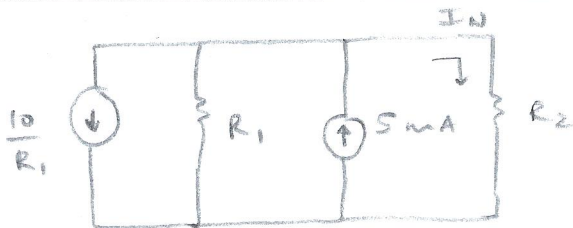


06 Determine I_N . [2]

07 Determine R_N . [2]

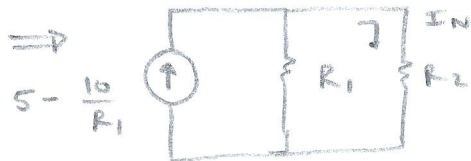
06

I_N :



$$R_1 = 7k\Omega$$

$$R_2 = 3k\Omega$$

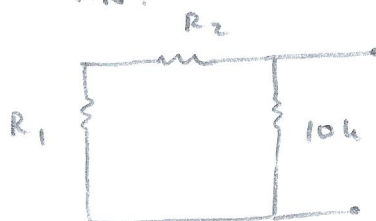


$$I_N = \left(5 - \frac{10}{R_1} \right) \left(\frac{R_1}{R_1 + R_2} \right)$$

$$I_N = \frac{5R_1 - 10}{R_1 + R_2} \text{ [mA]} \quad (*)$$

$$I_N = 3.572 \left(\frac{7k}{10k} \right) = 2.5 \text{ mA}$$

07: R_N :



$$R_1 = 7k$$

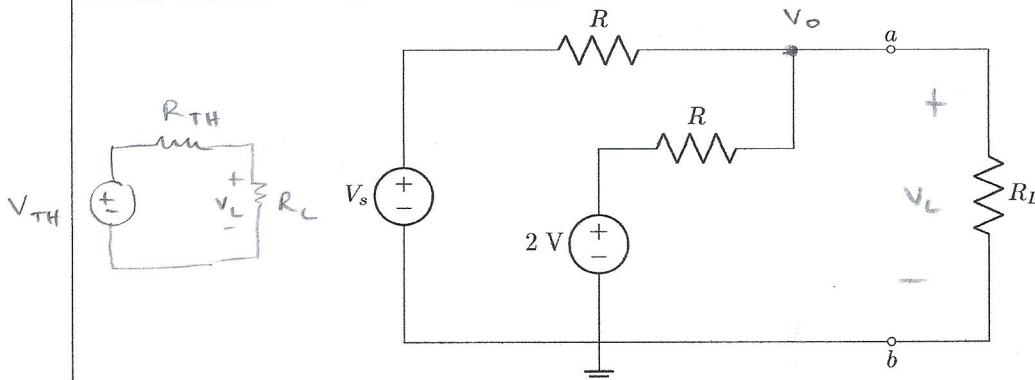
$$R_2 = 3k$$

$$R_N = (R_1 + R_2) \parallel 10k$$

$$R_N = \frac{10(R_1 + R_2)}{(10 + R_1 + R_2)} \text{ [k}\Omega\text{]} \quad (*)$$

$$R_N = \frac{10(10)}{20} = 5 \text{ k}\Omega$$

Questions 8 to 9 [4]



- 08 Determine the voltage V_L across the load R_L for maximum power transfer if $R = 250 \Omega$. [2]
(The value of V_s is given in ClickUP.)
- 09 Determine the value of R such that the maximum power P_{max} is delivered to the load, given that $V_s = V_L = 2 \text{ V}$. [2]
(The value of P_{max} is given in ClickUP.)

08:
$$\frac{V_o - V_s}{R} + \frac{V_o - 2}{R} = 0$$

$$2V_o = V_s + 2$$

$$V_o = \frac{V_s + 2}{2} = V_{TH}$$

$$V_L = \frac{V_{TH}}{2} \therefore V_L = \frac{V_s + 2}{4} \text{ [V]}$$

(max power $R_{TH} = R_L$)

$V_s = 4 \text{ V}$

$V_L = 1.5 \text{ V}$

09:
$$P_{max} = \frac{V_{TH}^2}{4R_{TH}} = \frac{(2V_L)^2}{4\left(\frac{R}{2}\right)} = 1 \text{ (} V_L = 1 \text{ V)}$$

$$P_{max} = 10 \text{ mW}$$

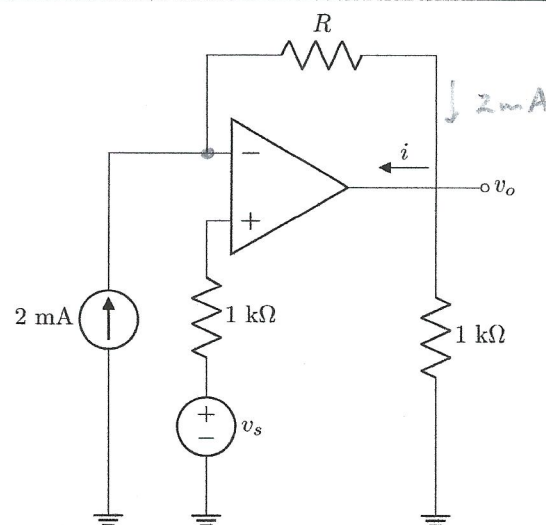
$$R_{TH} = \frac{R}{2}$$

$$P_{max} = \frac{4}{4\left(\frac{R}{2}\right)} = \frac{2}{R}$$

$$R = \frac{2}{P_{max}}$$

$$R = \frac{2}{10 \text{ mW}} = 200 \Omega$$

Question 10 [2]



- 10 Find i . [2]
(The values of v_s and R are given in ClickUP.)

$$V_s = 6V$$

$$R = 10k\Omega$$

$$-2mA + \frac{V_s - V_o}{R} = 0$$

$$V_o = V_s - 2mA(R)$$

$$i = 2mA - \frac{V_o}{1k\Omega}$$

$$i = 2mA - \frac{V_s - 2mA(R)}{1k\Omega}$$

$$i = 2 - \frac{V_s + 2R}{1k} [mA] (*)$$

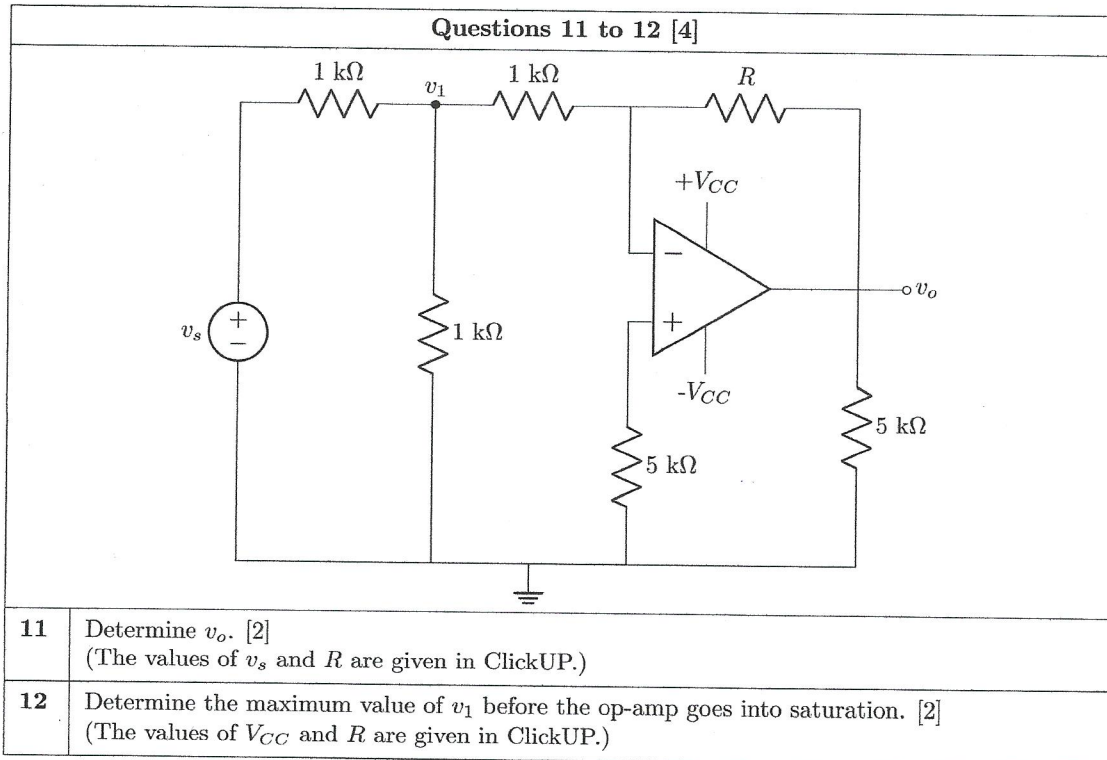
$$-2 + \frac{6 - V_o}{10} = 0$$

$$-20 + 6 = V_o$$

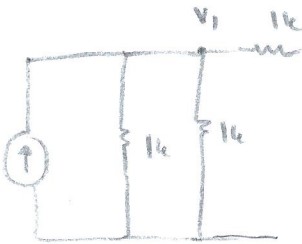
$$V_o = -14$$

$$i = 2 - \left(\frac{-14}{1k} \right)$$

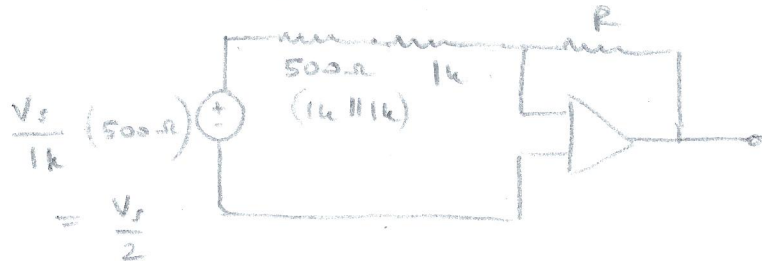
$$i = 16mA$$

11

$$\frac{V_s}{1k}$$



inverting Configuration:



$$v_o = -\frac{V_s}{2} \left(\frac{R}{1.5} \right)$$

$$R = 10k \Omega$$

$$V_s = 3$$

$$v_o = -\frac{V_s R}{3} \quad [V] \quad (*)$$

$$v_o = -\frac{3}{2} \left(\frac{10}{1.5} \right) = -10V$$

12

$$-V_{CC} \leq v_o \leq V_{CC}$$

$$R = 2k \Omega$$

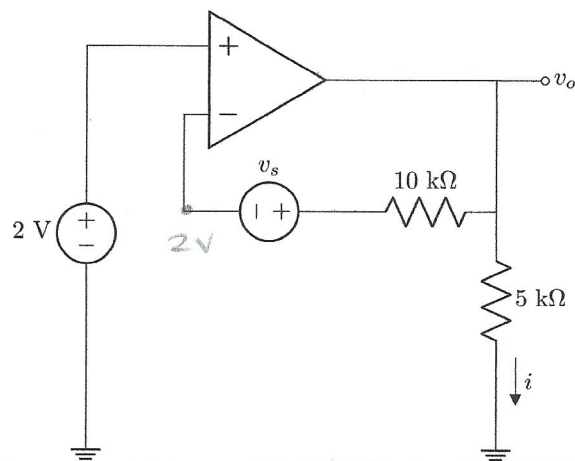
$$V_{CC} = 12V$$

$$-V_{CC} \leq -\frac{R}{1k} v_1 \leq V_{CC}$$

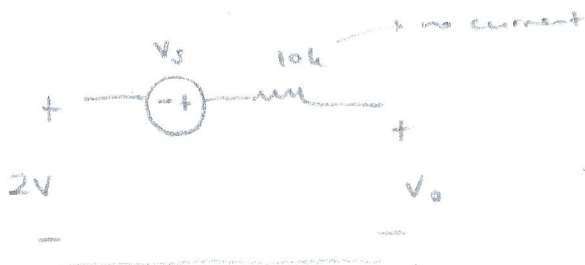
$$v_1 = \frac{V_{CC}}{R} \quad [V] \quad (*) \quad \text{max}$$

$$v_1 = \frac{12}{2} = 6V$$

Question 13 [2]



- 13 Find i . [2]
(The value of v_s is given in ClickUP.)



$$v_s = 3V$$

$$v_o = 2 + 3 = 5$$

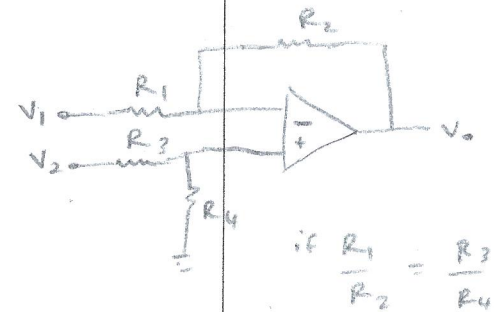
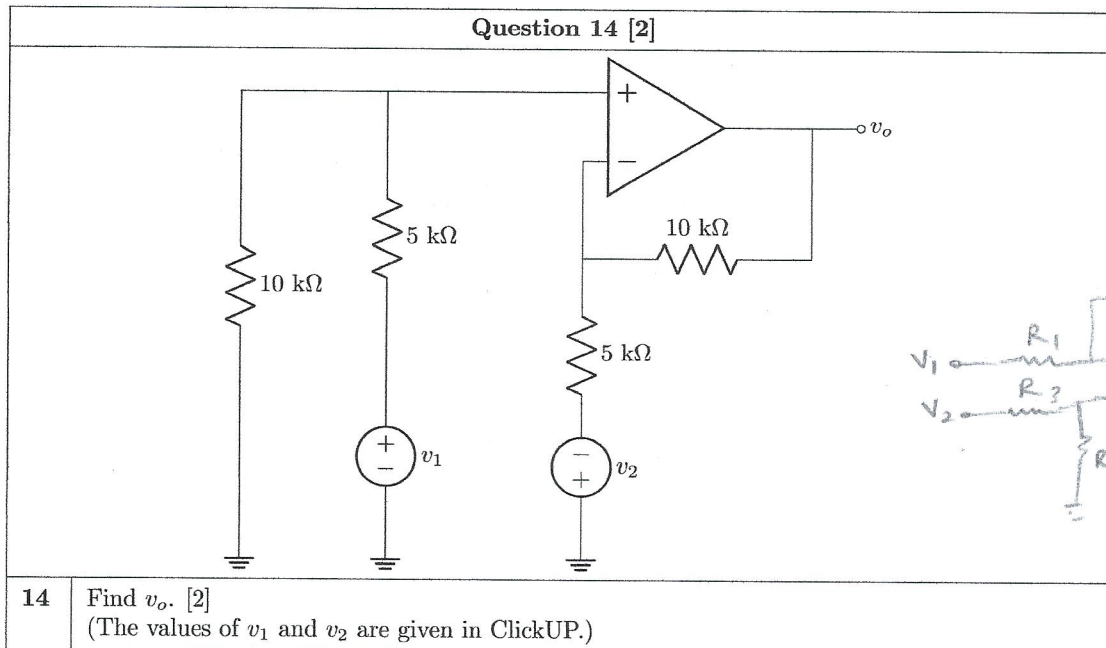
$$i = \frac{5}{5} = 1mA$$

$$-2V - v_s + v_o = 0$$

$$v_o = 2 + v_s$$

$$i = \frac{v_o}{5k} = \frac{2 + v_s}{5} [mA] *$$

Question 14 [2]



$$V_o = \frac{R_2}{R_1} (V_2 - V_1)$$

$$\Rightarrow V_o = 2 (V_1 - (-V_2))$$

difference amplifier:

$$V_o = \frac{R_2}{R_1} (V_1 - (-V_2)) \text{ as } \frac{R_1}{R_2} = \frac{R_3}{R_4}$$

$$V_o = \frac{10k}{5k} (V_1 + V_2)$$

$$V_o = 2 (V_1 + V_2) [V] (*)$$

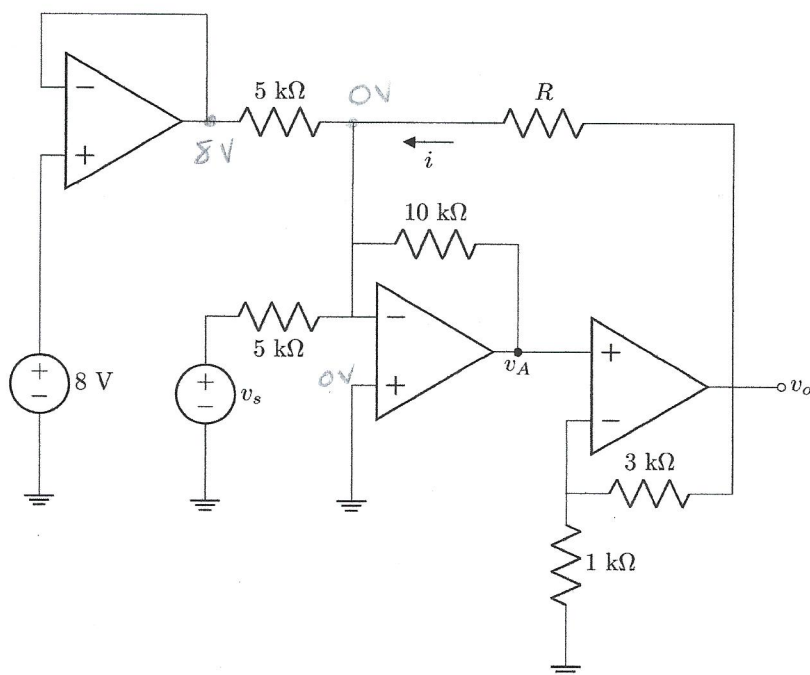
$$\text{if } V_1 = 3V$$

$$V_2 = 2V$$

$$V_o = 10V$$



Questions 15 to 16 [4]



- 15 Determine v_A when $R \rightarrow \infty$. [2]
(The value of v_s is given in ClickUP.)
- 16 Determine i when $v_s = 6$ V. [2]
(The values of R and v_A are given in ClickUP.)

15 summing amp: $V_A = - \left(\frac{10k}{5k} v_s + \frac{10k}{5k} (8) \right)$ $v_s = 6V$

$$V_A = - (2v_s + 16)$$

$$V_A = -2(v_s + 8) [V] \text{ *}$$

$$V_A = - (2(6) + 16)$$

$$V_A = -28V$$

16 non-inverting: $V_o = \left(1 + \frac{3k}{1k} \right) V_A$

and $i = \frac{V_o - 0}{R} \therefore i = \frac{(1+3)V_A}{R} = \frac{4V_A}{R} [mA] \text{ *}$

Total Section 1 : [32]

Grand Total: [32]

$V_A = 15V$

$R = 2k\Omega$

$$V_o = 4(15)$$

$$= 60$$

$$i = \frac{60}{2} = 30mA$$